```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score

df = pd.read_csv("body_weight_dataset.csv")

df
```

```
     Weight  Height  Age  Gender
0      67.5    1.50   41    Male
1      61.9    1.58   31  Female
2      99.8    1.95   58    Male
3      73.0    1.72   46  Female
4      92.9    1.89   41    Male
..      ...     ...  ...     ...
195    81.1    1.77   52    Male
196    69.1    1.67   26    Male
197    70.4    1.69   53  Female
198    76.6    1.79   24  Female
199    80.5    1.75   40    Male

[200 rows x 4 columns]
```

```
df.head()
```

```
   Weight  Height  Age  Gender
0    67.5    1.50   41    Male
1    61.9    1.58   31  Female
2    99.8    1.95   58    Male
3    73.0    1.72   46  Female
4    92.9    1.89   41    Male
```

```
df.tail()
```

```
     Weight  Height  Age  Gender
195    81.1    1.77   52    Male
196    69.1    1.67   26    Male
197    70.4    1.69   53  Female
198    76.6    1.79   24  Female
199    80.5    1.75   40    Male
```

```
df.shape
```

```
(200, 4)
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Weight  200 non-null    float64
 1   Height  200 non-null    float64
 2   Age     200 non-null    int64
 3   Gender  200 non-null    str
dtypes: float64(2), int64(1), str(1)
memory usage: 6.4 KB
```

```
df.isnull().sum()
```

```
Weight    0
Height    0
Age       0
Gender    0
dtype: int64
```

```
df.describe()
```

	Weight	Height	Age
count	200.00000	200.000000	200.0000
mean	74.59600	1.697600	42.2350
std	11.09083	0.141294	13.7135
min	50.50000	1.450000	18.0000
25%	66.47500	1.580000	31.0000
50%	74.95000	1.690000	42.5000
75%	83.50000	1.810000	55.0000
max	99.80000	1.950000	65.0000

```
df.duplicated().sum()
```

```
np.int64(1)
```

```
df["Gender"] = df["Gender"].map({
    "Male":1,
    "Female":0
})
```

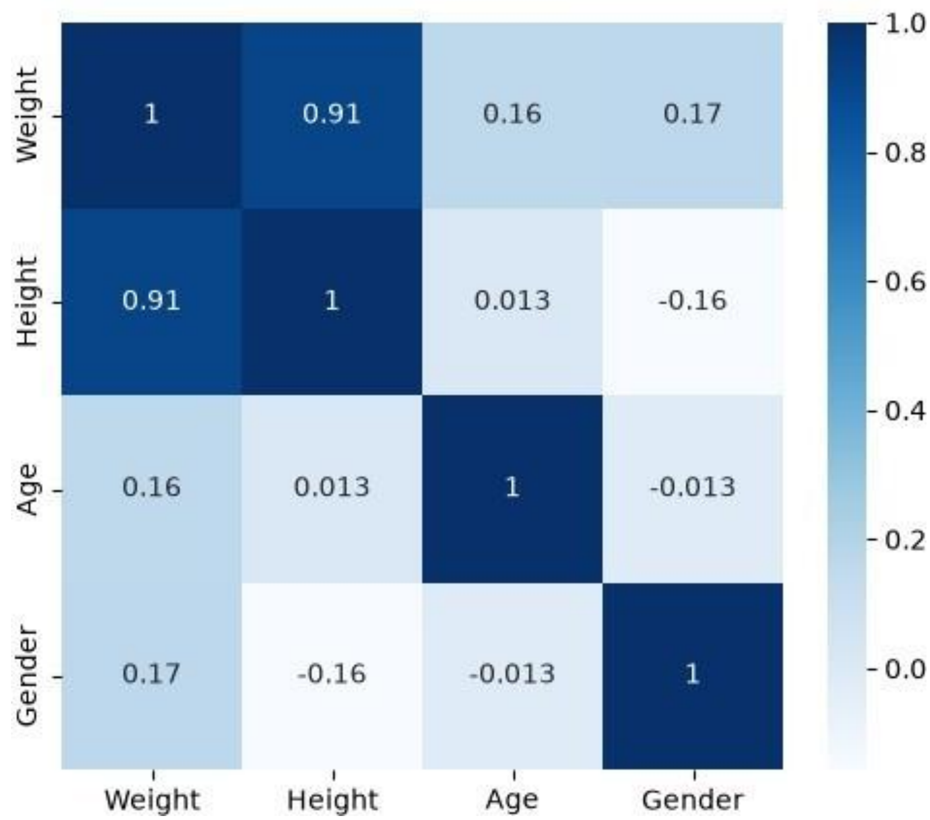
```
df.head()
```

	Weight	Height	Age	Gender
0	67.5	1.50	41	1
1	61.9	1.58	31	0
2	99.8	1.95	58	1
3	73.0	1.72	46	0
4	92.9	1.89	41	1

```
df.corr()
```

	Weight	Height	Age	Gender
Weight	1.000000	0.911290	0.162013	0.166781
Height	0.911290	1.000000	0.012974	-0.156432
Age	0.162013	0.012974	1.000000	-0.013413
Gender	0.166781	-0.156432	-0.013413	1.000000

```
plt.figure(figsize=(6,5))
sns.heatmap(df.corr(),annot=True,cmap="Blues")
plt.show()
```




```
plt.hist(df["Weight"],bins=15)
plt.title("Weight Distribution")
plt.show()
```

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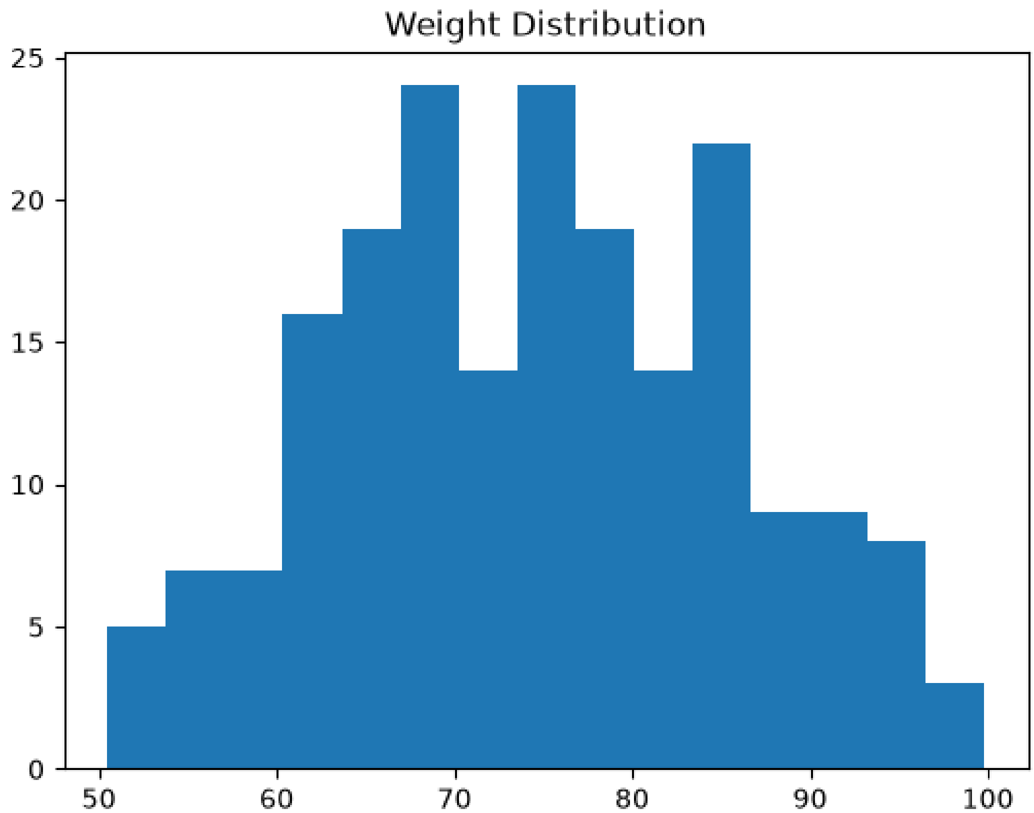
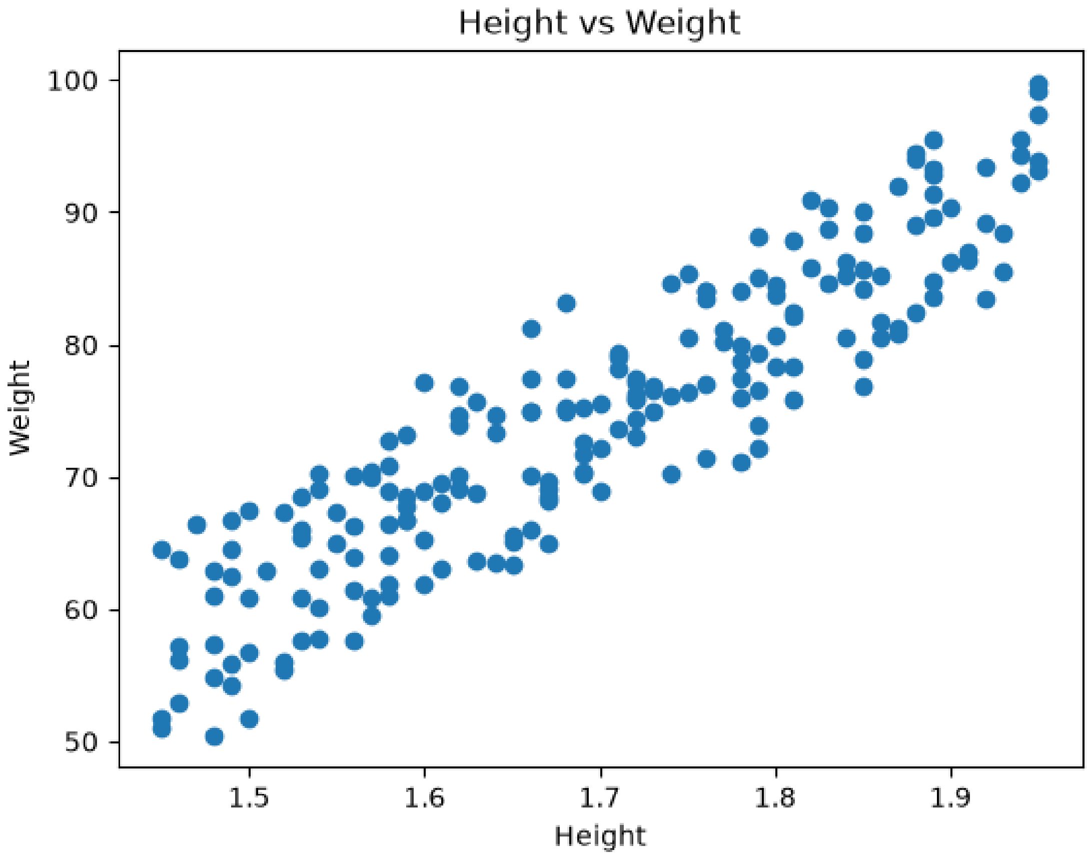
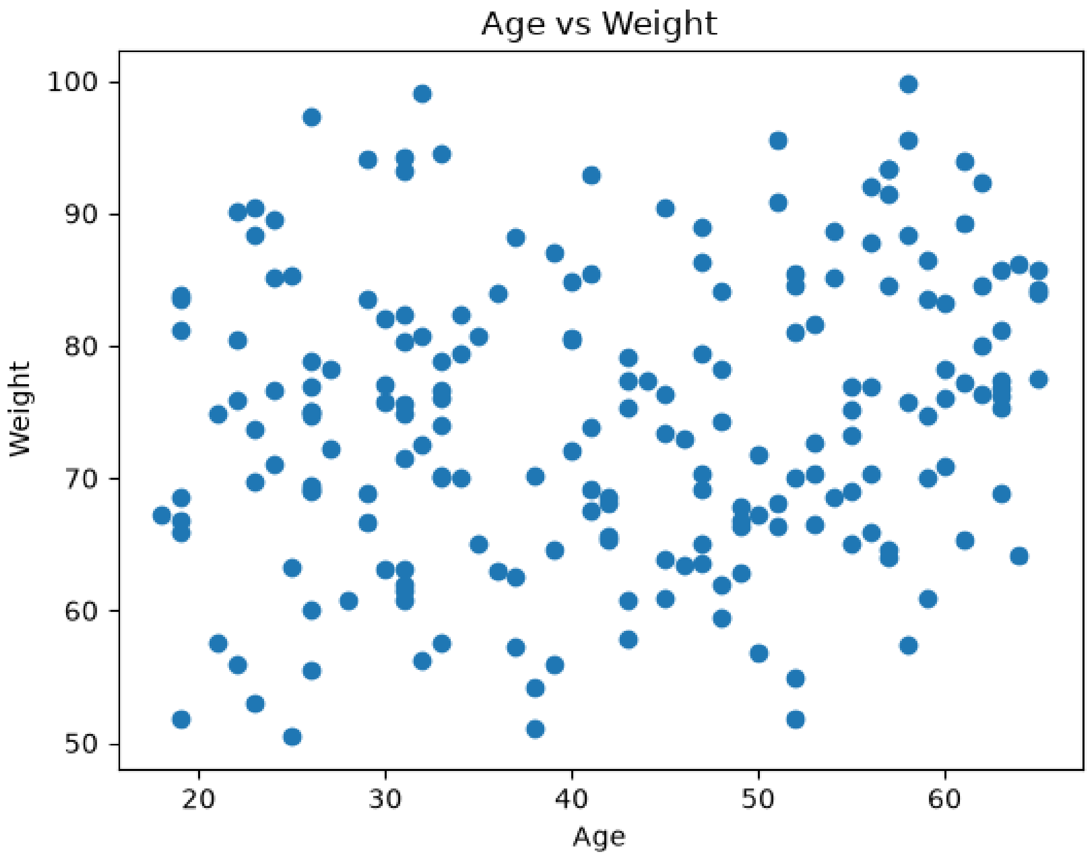


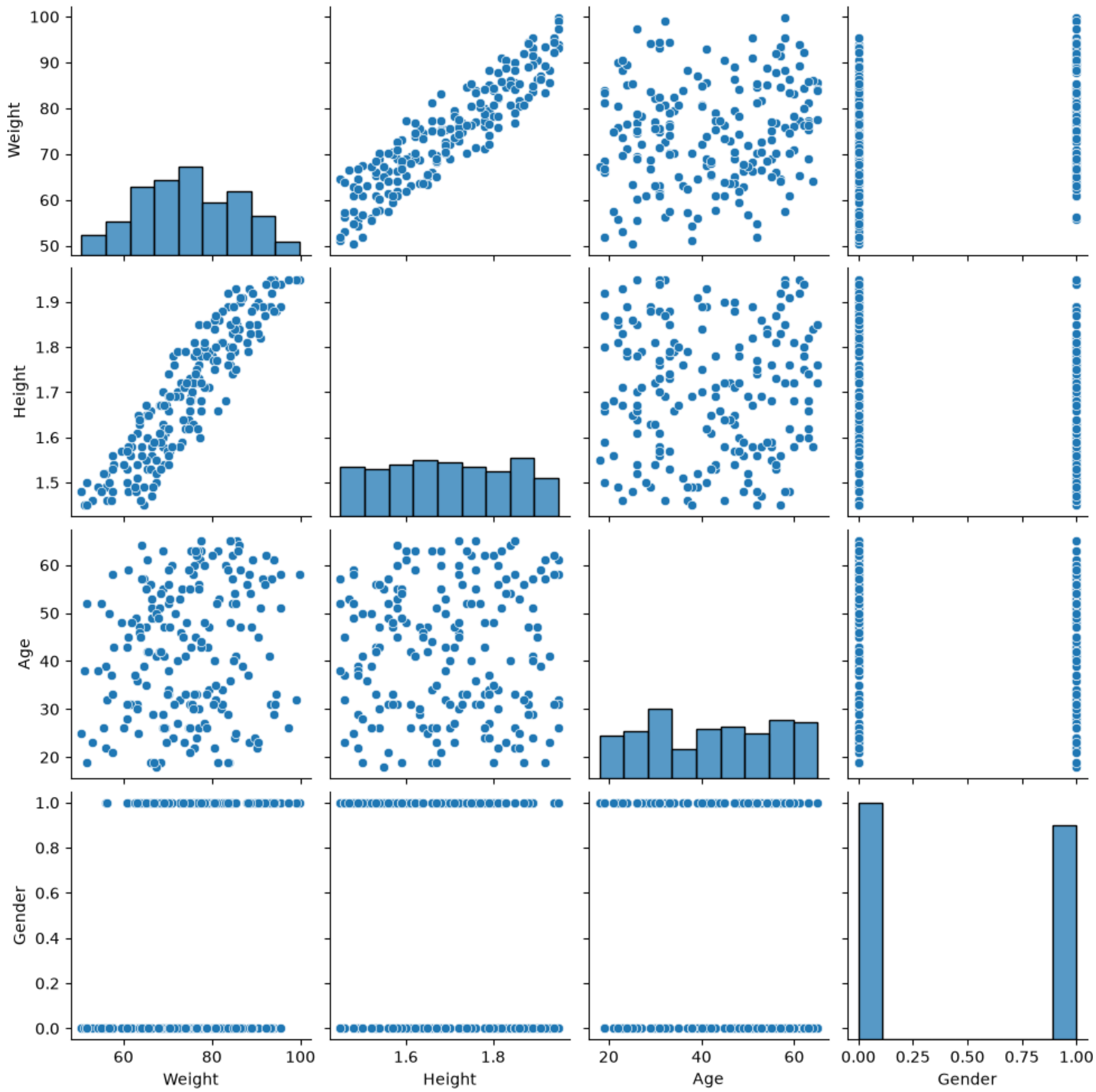

```
plt.scatter(df["Height"],df["Weight"])
plt.xlabel("Height")
plt.ylabel("Weight")
plt.title("Height vs Weight")
plt.show()
```




```
plt.scatter(df["Age"],df["Weight"])
plt.xlabel("Age")
plt.ylabel("Weight")
plt.title("Age vs Weight")
plt.show()
```




```
sns.pairplot(df)
plt.show()
```




```python
X = df[["Height","Age","Gender"]]

y = df["Weight"]

X.head()
```

	Height	Age	Gender
0	1.50	41	1
1	1.58	31	0
2	1.95	58	1
3	1.72	46	0
4	1.89	41	1

```
y.head()
```

```
0    67.5
1    61.9
2    99.8
3    73.0
4    92.9
Name: Weight, dtype: float64
```

```
X_train,X_test,y_train,y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

```
print(X_train.shape)
print(X_test.shape)
```

```
(160, 3)
(40, 3)
```

```
model = LinearRegression()
```

```
model.fit(X_train,y_train)
```

```
LinearRegression()
```

```
y_pred = model.predict(X_test)
```

```
print("MAE =",mean_absolute_error(y_test,y_pred))
```

```
MAE = 2.0998709048547246
```

```
mse = mean_squared_error(y_test,y_pred)
print("MSE =",mse)
```

```
MSE = 6.329184781005528
```

```
print("RMSE =",np.sqrt(mse))
```

```
RMSE = 2.51578710963498
```

```
print("R2 Score =",r2_score(y_test,y_pred))
```

```
R2 Score = 0.9552151618608655
```

```
print(model.coef_)
```

```
[76.17577728  0.11976253  7.28333164]
```

```
print(model.intercept_)
```

```
-63.216848278290186
```

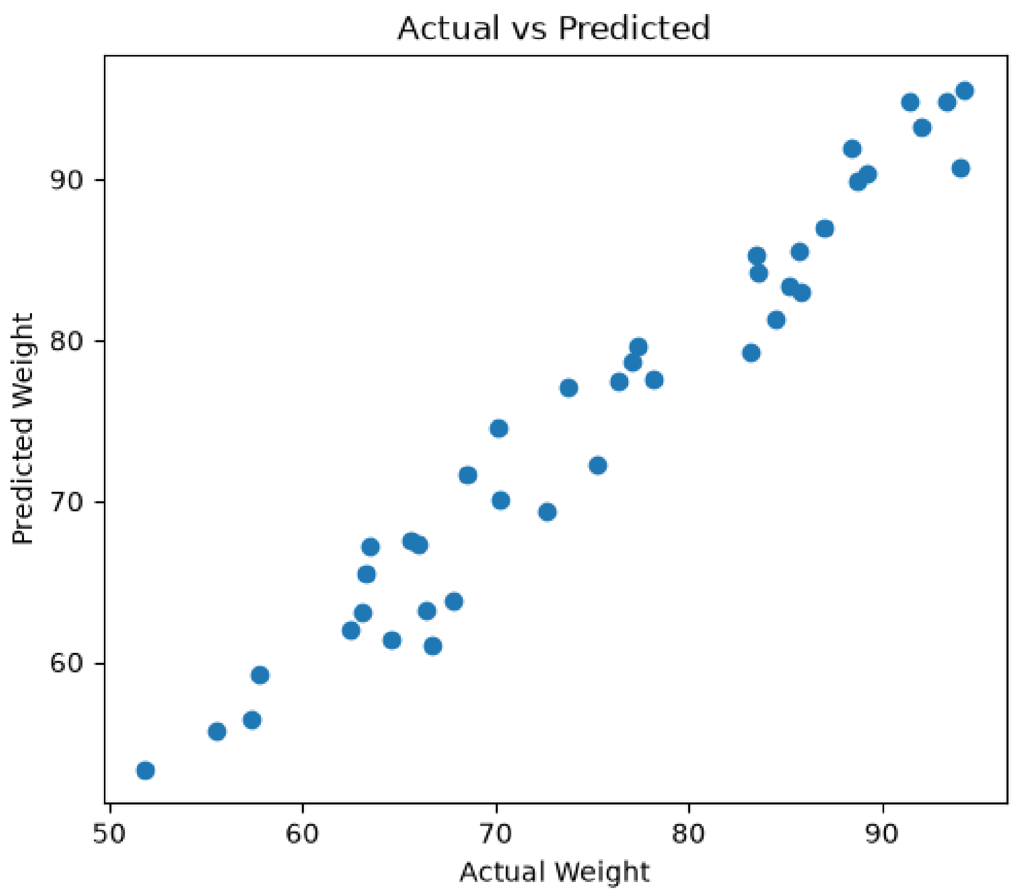
```
person = pd.DataFrame({
    "Height":[1.75],
    "Age":[25],
    "Gender":[1]       # Male=1, Female=0
})

prediction = model.predict(person)

print("Predicted Weight =",prediction[0],"kg")
```

```
Predicted Weight = 80.3681568945941 kg
```

```
results = pd.DataFrame({
    "Actual":y_test,
    "Predicted":y_pred
})

results.head()
```

	Actual	Predicted
95	57.8	59.243638
15	72.6	69.352616
30	64.6	61.347825
158	92.0	93.221889
128	94.1	90.750058

```
plt.figure(figsize=(6,5))

plt.scatter(y_test,y_pred)

plt.xlabel("Actual Weight")

plt.ylabel("Predicted Weight")

plt.title("Actual vs Predicted")

plt.show()
```




```
results.to_csv("BodyWeightPrediction.csv",index=False)

print("Prediction file saved successfully.")
```

```
Prediction file saved successfully.
```